Hierarchical POMDP: federated belief sharing across levels

The flat sentinel world couples all agents at a single latent level — the creature's location.

Hierarchical POMDP (part 2)

A natural extension is a **2-level hierarchical POMDP** in which location inference (Level 1, L1; 9 states) is coupled to a global *context* variable (Level 2, L2; 2 states: `quiet` / `alert`) that modulates the L1 prior.

Hierarchical POMDP (part 3)

In the `alert` context the creature is expected near the den (center cell); in the `quiet` context the prior is uniform.

Hierarchical POMDP (part 4)

Each sentinel runs alternating L1/L2 minimization with `hierarchical_infer` from the `fedference.pomdp` module to infer both its location belief and the current context belief, then the colony federates both levels via a log-linear pool.

Hierarchical POMDP (part 5)

We compare two conditions over 960 seeded trials with 4 agents at sensor acuity 0.85:

Hierarchical POMDP (part 6)

- **Flat** — agents ignore the hierarchy and infer location under a uniform prior;
- **Hierarchical** — agents run 4 alternating-minimization iterations to couple L1 and L2 beliefs before federating.

Hierarchical POMDP (part 7)

The measured location accuracies are 0.979 (flat) and 0.969 (hierarchical), a gap of -0.010.

Hierarchical POMDP (part 8)

Across 128 independent seeds the hierarchical location accuracy is 0.974 (SD 0.005; 95 % CI 0.973–0.975) versus flat 0.984 (95 % CI 0.983–0.985), a mean accuracy gap of -0.010 (95 % CI -0.011—0.009; Wilcoxon signed-rank $p = 9.45 \times 10^{-23}$, effect size $r = 0.999$, large).

Hierarchical POMDP (part 9)

On location the gap is small but statistically reliable in the *negative* direction — the paired test rejects at $\alpha = 0.05$ and the gap's confidence interval (-0.011—0.009) excludes zero on the negative side — so the hierarchy does not improve location accuracy in this regime; if anything it pays a small,

Hierarchical POMDP (part 10)

consistent location cost for carrying the extra latent level.

Hierarchical POMDP (part 11)

Its added value is that it *also* infers the context latent, at accuracy 0.763 against a two-state chance baseline of 0.5. Two-level federation therefore runs L1/L2 inference end-to-end and resolves context above chance while paying a small, reliable location cost relative to the flat baseline (fig. 31).

Hierarchical POMDP (part 12)

Context beliefs across the alternating-minimization iterations are shown in the top-middle panel: P(alert) sits above the two-state chance line and is stable from the first iteration onward when the observed location is the center cell —

Hierarchical POMDP (part 13)

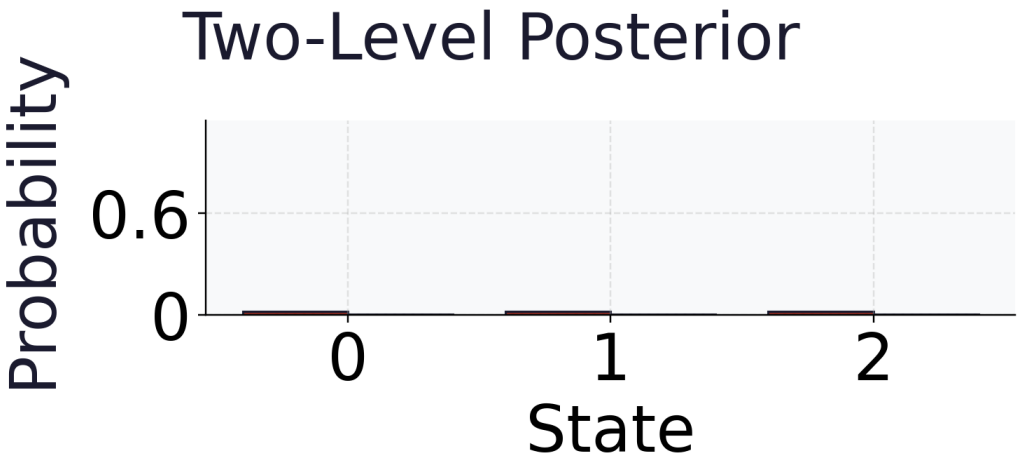
the center-cell observation pins the context posterior immediately, because the alert context-conditioned L1 prior is peaked there.

Hierarchical POMDP (part 14)

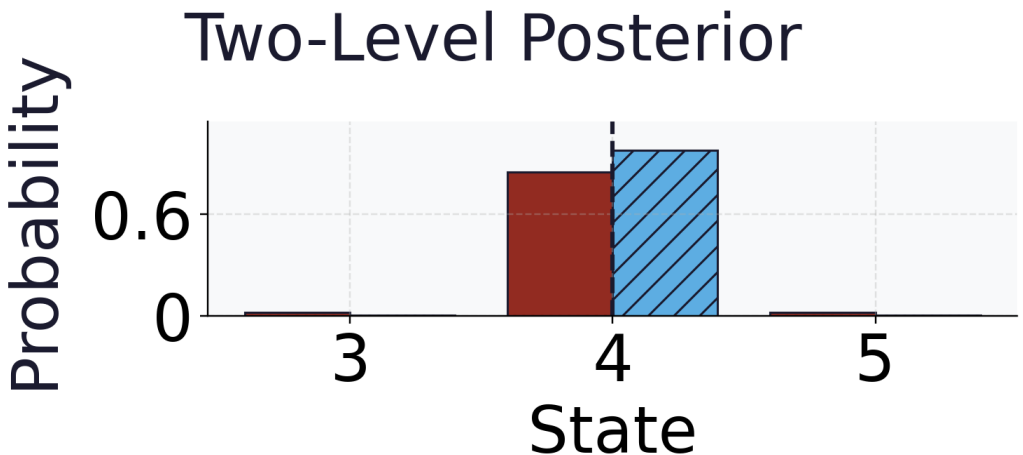
The full construction and parameter sweep are detailed in the supplement (sec. 17.10). For the effect of acuity and colony size on these results, see sec. 17.13.

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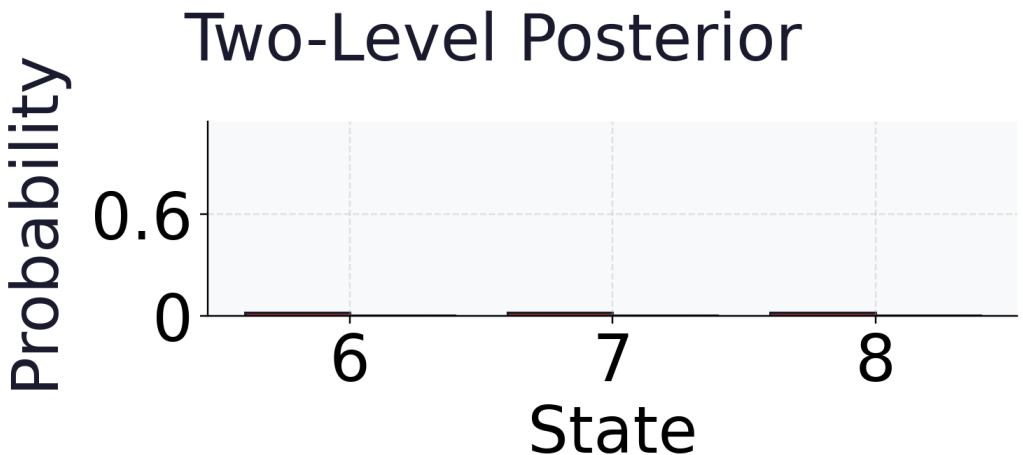
Hierarchical POMDP (part 15)



Hierarchical POMDP (part 16)



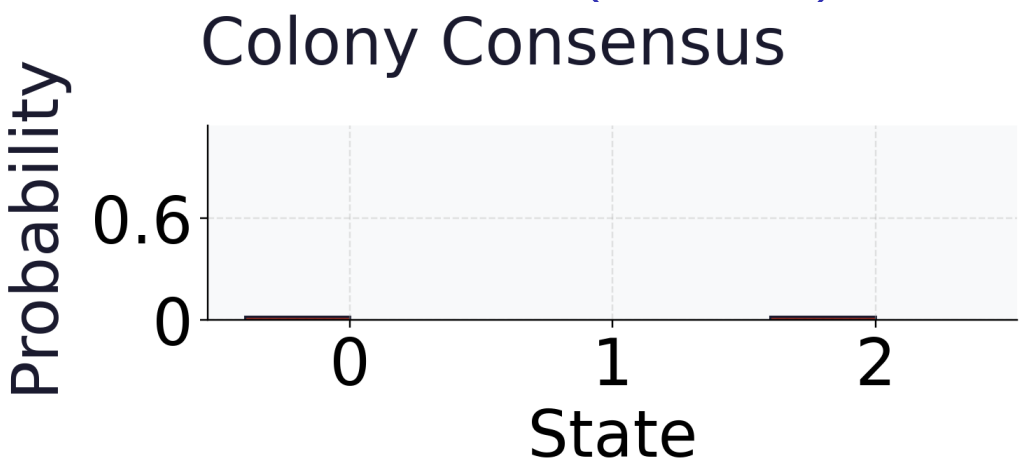
Hierarchical POMDP (part 17)



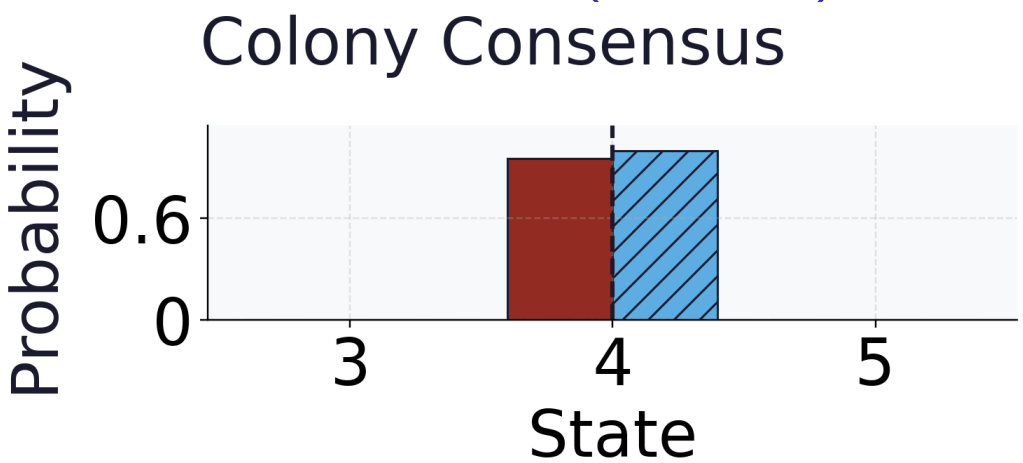
Hierarchical POMDP (part 18)

Two-Level Posterior: flat peak state 4, mass 0.85; hierarchical peak state 4, mass 0.9795.

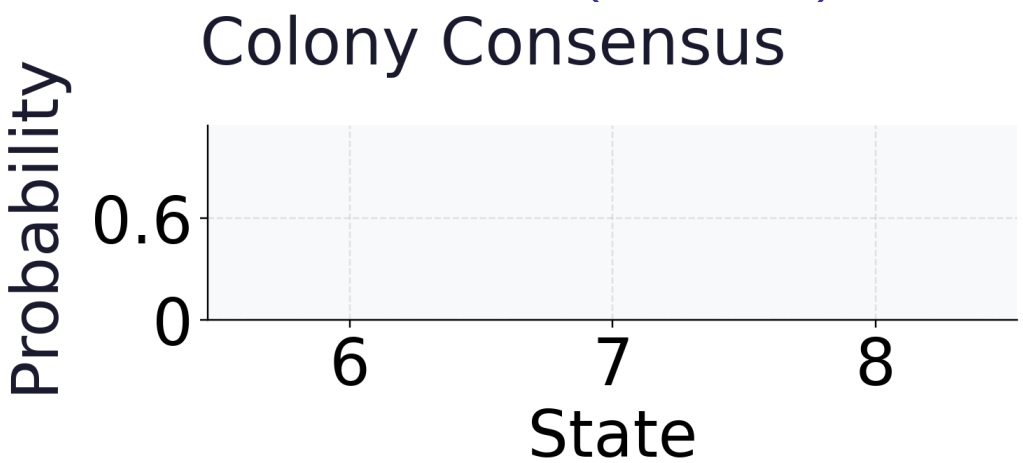
Hierarchical POMDP (part 19)



Hierarchical POMDP (part 20)



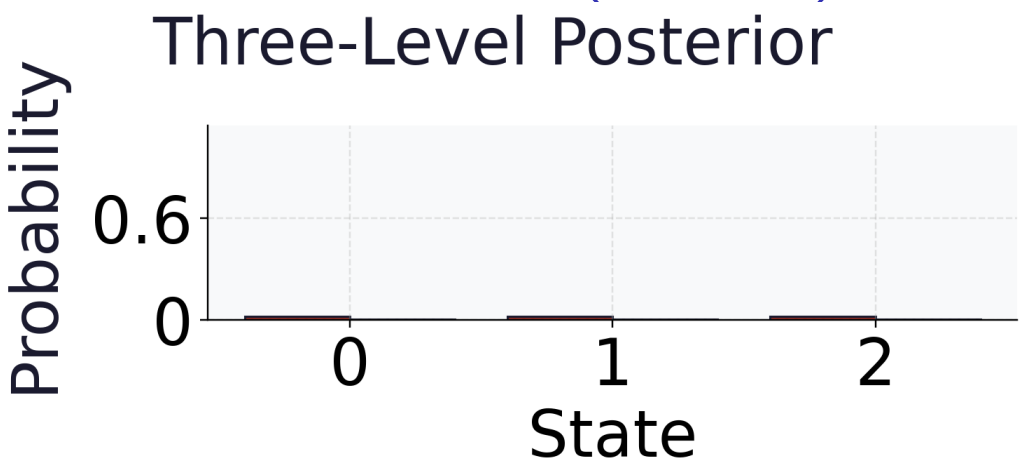
Hierarchical POMDP (part 21)



Hierarchical POMDP (part 22)

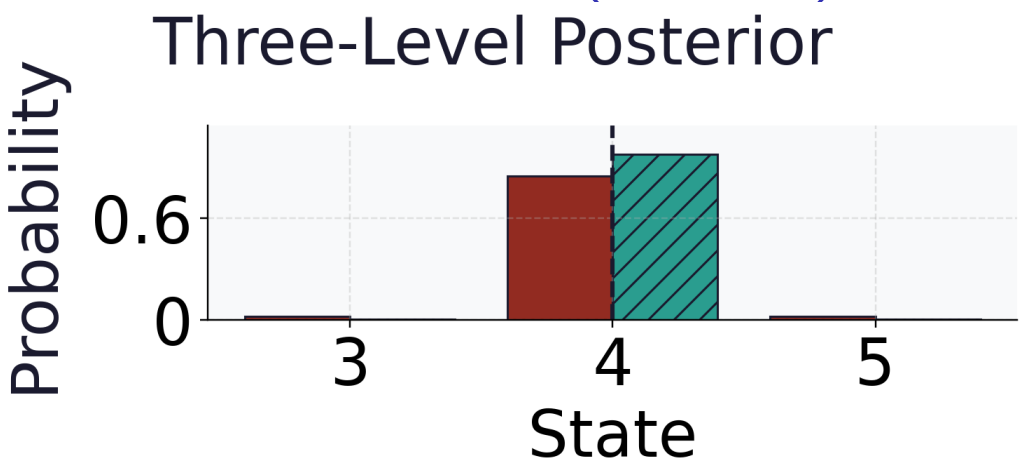
Colony Consensus: flat peak state 4, mass 0.9551; hierarchical peak state 4, mass 0.9999.

Hierarchical POMDP (part 23)

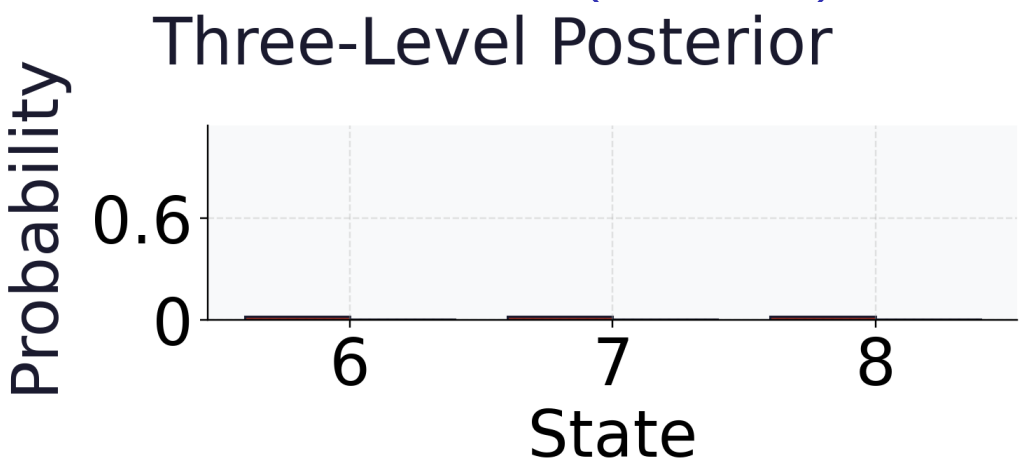


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Hierarchical POMDP (part 24)



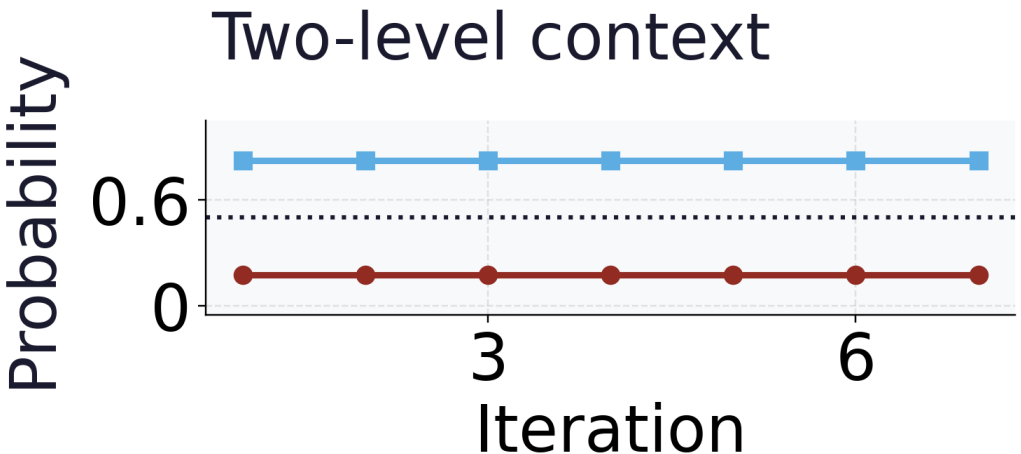
Hierarchical POMDP (part 25)



Hierarchical POMDP (part 26)

Three-Level Posterior: flat peak state 4, mass 0.85; hierarchical peak state 4, mass 0.9802.

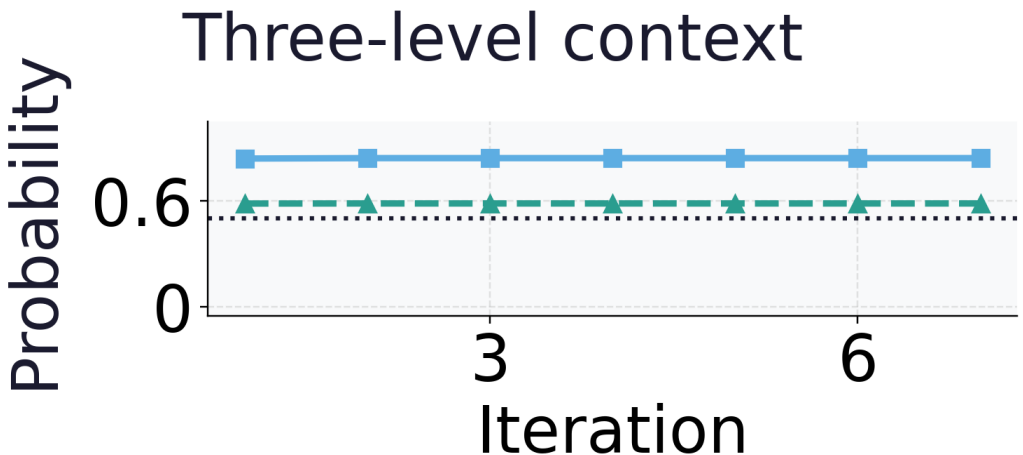
Hierarchical POMDP (part 27)



Hierarchical POMDP (part 28)

Two-level context: quiet (o, -); alert (s, -). Ordered iterations are not independent replications.

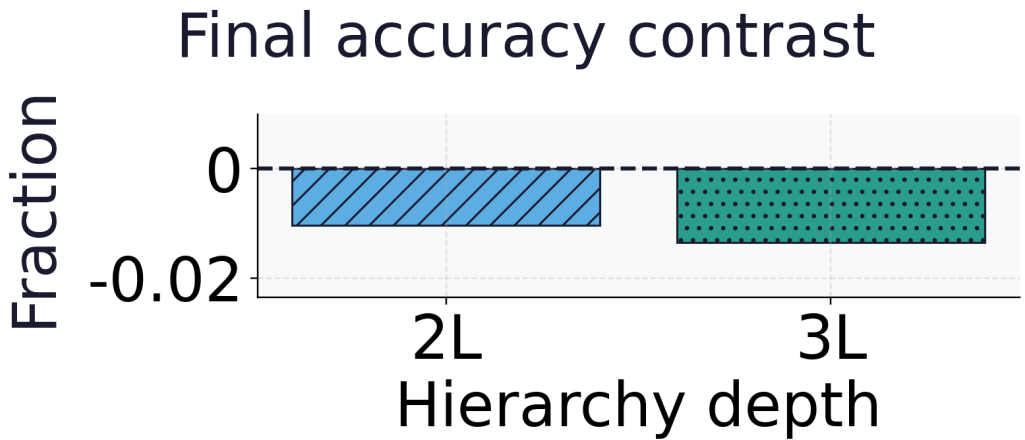
Hierarchical POMDP (part 29)



Hierarchical POMDP (part 30)

Three-level context: L2 alert (s, -); L3 high threat (^, --). Ordered iterations are not independent replications.

Hierarchical POMDP (part 31)



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Hierarchical POMDP (part 32)

Hierarchical minus flat: 2L -0.01042; 3L -0.01354. n = 960 nested trials. No per-trial trajectory or interval is inferred from these scalar gaps.

Hierarchical POMDP (part 33)

Acuity 0.85; observed state 4; colony true state 4; 4 agents. Plain left bars: flat; hatched right bars: hierarchical.

Hierarchical POMDP (part 34)

Gap bars use the executed reports. Other panels are seeded diagnostic posterior calculations. No universal hierarchical benefit or calibrated uncertainty is established.